

**LOOPS**

# PRINT FROM 1 TO 5

```
#include<iostream>
using namespace std;

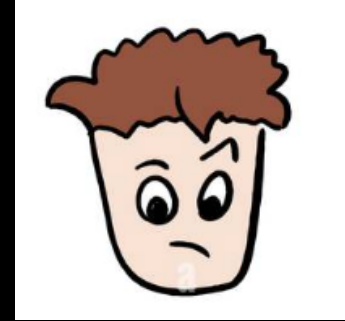
int main()
{
    cout << 1 << endl;
    cout << 2 << endl;
    cout << 3 << endl;
    cout << 4 << endl;
    cout << 5 << endl;
}
```

No double quote  
gpl



# PRINT FROM 1 TO 10

```
int main()
{
    cout << 1 << endl;
    cout << 2 << endl;
    cout << 3 << endl;
    cout << 4 << endl;
    cout << 5 << endl;
    cout << 6 << endl;
    cout << 7 << endl;
    cout << 8 << endl;
    cout << 9 << endl;
    cout << 10 << endl;
}
```



# PRINT FROM 1 TO 1000

```
int main()
{
    cout << 1 << endl;
    cout << 2 << endl;
    cout << 3 << endl;
    cout << 4 << endl;
    cout << 5 << endl;
    cout << 6 << endl;
    cout << 7 << endl;
    cout << 8 << endl;
    cout << 9 << endl;
    cout << 10 << endl;
```

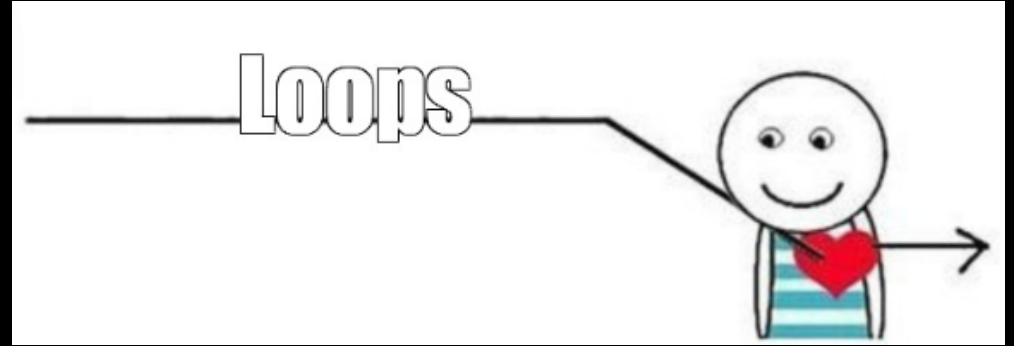
·  
·  
·  
·  
·



How can we repeat work without  
writing the same code again and  
again?

# LOOPS

When you want to  
repeat an action  
again and again  
till a certain condition,  
we can use loops.



# TYPE OF LOOPS

1) **While** loop

2) **For** loop

3) **Do-while** loop

# WHILE LOOP

```
initialization;
```

```
while(condition)
```

```
{
```

```
    // body
```

```
    update;
```

```
}
```

# PRINT FROM 1 TO 10

```
int i = 1; False
while(i <= 10)
{
    cout << i << endl;
    i++;
}
```

1  
2  
3  
4  
5

6  
7  
8

9  
10

11 <= 10 *No*

i

1 2 3 4 5 6 7 8 9 10 11

1 <= 10 YES

2 <= 10 YES

3 <= 10 YES

4 <= 10 YES

5 <= 10 YES



# PRINT 1 TO N

```
int main()
{
    int n;
    cin >> n;

    int i = 1;
    while(i <= n)
    {
        cout << i << endl;
        i++;
    }
}
```

]

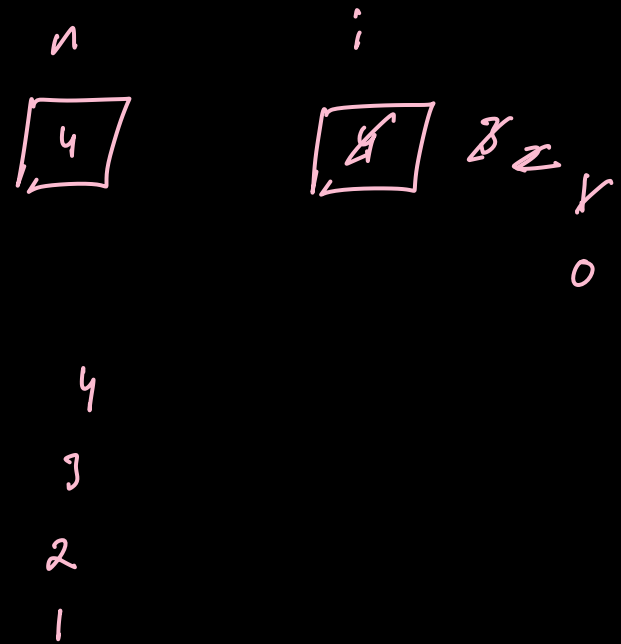


```
int i = n;
while ( i >= 1)
{
    cout << i << endl;
    i--;
}
```

# PRINT N TO 1

```
int main()
{
    int n;
    cin >> n;

    int i = n;
    while(i >= 1)
    {
        cout << i << endl;
        i--;
    }
}
```



# PRINT NUMBERS FROM L TO R

```
int main()
{
    int l, r;
    cin >> l >> r;

    int i = l;
    while(i <= r)
    {
        cout << i << endl;
        i++;
    }
}
```

# PRINT ALL EVEN NUMBERS FROM 1 TO N

n = 5

2

4

n = 6

2

4

6

```
int n;  
cin >> n;  
  
int i = 1;  
while(i <= n)  
{  
    if(i % 2 == 0)  
    {  
        cout << i << endl;  
    }  
    i++;  
}
```

| i                       | n                       |
|-------------------------|-------------------------|
| <div><div>✓</div></div> | <div><div>5</div></div> |
| 2                       |                         |
| 3                       |                         |
| 4                       |                         |
| 5                       |                         |
| 6                       | 2                       |
|                         | 4                       |

# PRINT ALL EVEN NUMBERS FROM 1 TO N

**n = 5**

2

4

**n = 6**

2

4

6

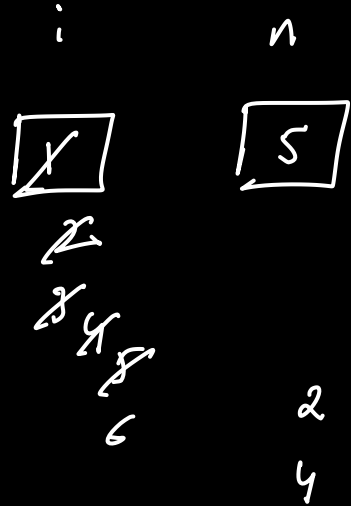
```
int n;  
cin >> n;
```

```
int i = 2;  
while(i <= n)  
{
```

```
    cout << i << endl;
```

```
    i += 2;
```

```
}
```



# PRINT ALL UPPERCASE ALPHABETS (A - Z)

```
char ch = 'A';
```

```
while (ch <= 'Z')
```

```
{
```

```
    cout << ch << endl;
```

```
    ch++;
```

```
}
```

ASCII Values

'A' → 65

'a' → 97

'B' → 66

'b' → 98

'C' → 67

'c' → 99

⋮

⋮

# PRINT MULTIPLICATION TABLE

**n = 5**

$$5 \times 1 = 5$$

$$5 \times 2 = 10$$

$$5 \times 3 = 15$$

$$5 \times 4 = 20$$

$$5 \times 5 = 25$$

$$5 \times 6 = 30$$

$$5 \times 7 = 35$$

$$5 \times 8 = 40$$

$$5 \times 9 = 45$$

$$5 \times 10 = 50$$

**n = 6**

$$6 \times 1 = 6$$

$$6 \times 2 = 12$$

$$6 \times 3 = 18$$

$$6 \times 4 = 24$$

$$6 \times 5 = 30$$

$$6 \times 6 = 36$$

$$6 \times 7 = 42$$

$$6 \times 8 = 48$$

$$6 \times 9 = 54$$

$$6 \times 10 = 60$$

```
int i=1;
```

```
while(i <= 10)
```

```
{
```

```
cout << n << " x " << i << " = " <<
```

```
i++;
```

```
}
```

$(n \times i)$  << endl;

# PRINT NUMBER IN REVERSE

**n = 512**

215

**n = 1927**

7291

**n = 3429**

9243

```
int n;  
cin >> n;
```

```
while ( n != 0 )
```

```
{
```

```
    cout << n % 10;
```

```
    n = n / 10;
```

```
}
```

lastDigit = n % 10

after division by 10, a number loses its last

n = 1950

n

1927

192

19

0



# DIGITS SUM

Sum

0

$$n = 512 \Rightarrow 5 + 1 + 2 = 8$$

8

$$n = 1927 \Rightarrow 1 + 9 + 2 + 7 = 19$$

19

$$n = 3429 \Rightarrow 3 + 4 + 2 + 9 = 18$$

18

```
int sum = 0;
```

DSA Contd

```
int n;
```

```
cin >> n;
```

28th Jan

Wed

8-10 PM

```
while ( n != 0 )
```

```
{
```

```
    sum += n % 10;
```

```
    n = n / 10;
```

```
}
```

```
cout << sum << endl;
```

# REVERSE AND STORE IN A VARIABLE

**n = 512**

ans = 215

**n = 1927**

ans = 7291

**n = 3429**

ans = 9243

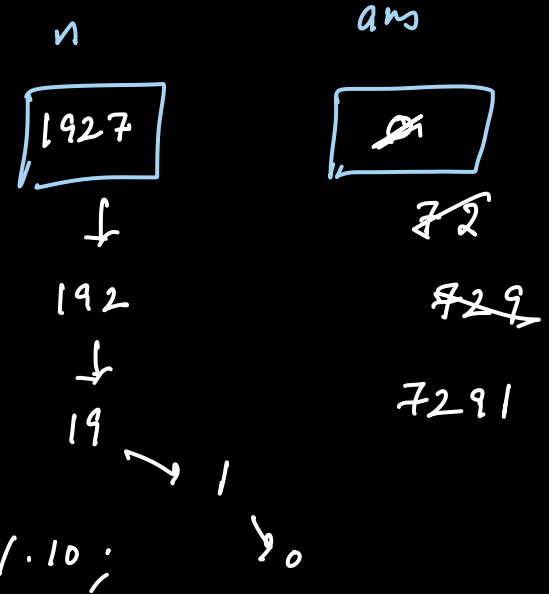
```
int ans = 0;

int n;
cin >> n;

while ( n != 0 )
{
    ans = (ans * 10) + n % 10;

    n = n / 10;
}

cout << ans << endl;
```



# PALINDROME

n = 12321

YES

rev = 12321

n = 1927

NO

rev = 7291

n = 1

YES

rev = 1

```
int rev = 0;
```

```
int n;
```

```
cin >> n;
```

```
int temp = n;
```

```
while ( n != 0 )
```

```
{    rev = (rev * 10) + n % 10;
```

```
    n = n / 10;
```

```
}
```

```
if ( rev == n ) → temp
```

```
{    cout << "YES";
```

```
}
```

```
else
```

```
{
```

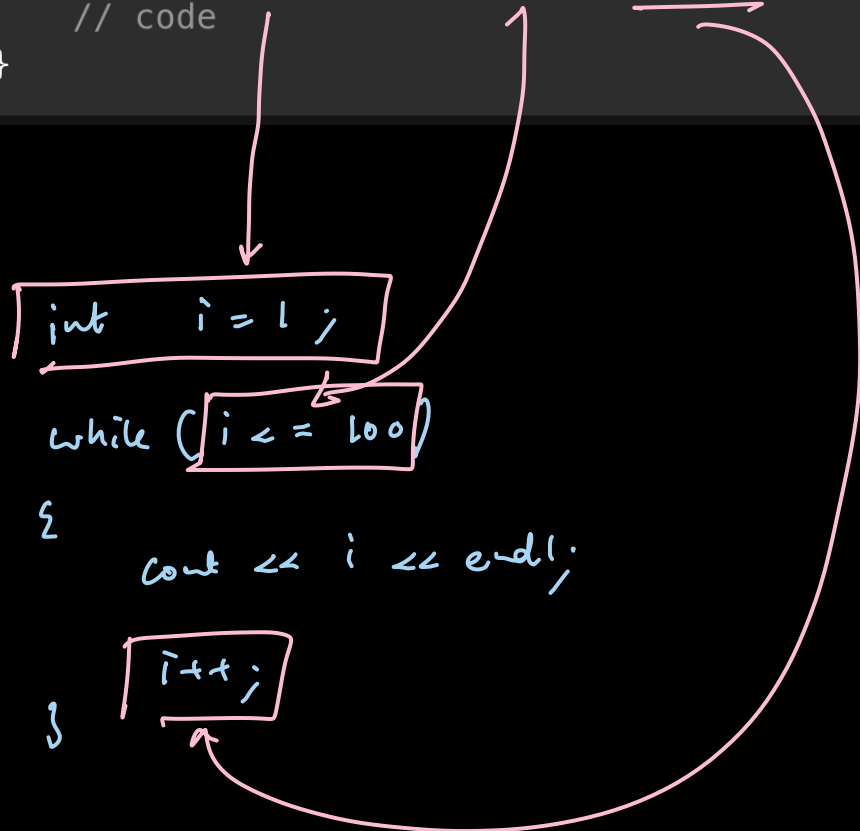
what will be the  
value of N?



# FOR LOOP

```
    cout << "No";  
}
```

```
for(initialization; condition; update) {  
    // code  
}
```



i  
1 2 3

```
for (int i=1; i <= 100; i++)
```

```
{
```

```
    cout << i << endl;
```

```
}
```

while ( )  for loop   




**PATTERN PRINTING**

# PRINT 1 TO N

Given  $n$  (number of rows), print the following pattern:

$n = 5$

1

2

3

4

5

```
for(i = 1; i <= n; i++)  
{  
    cout << i << endl;  
}
```

$n = 6$

1

2

3

4

5

6

$n = 7$

1

2

3

4

5

6

7



# 2 STARS

Given  $n$  (number of rows), print the following pattern:

$n = 5$

\*\*

\*\*

\*\*

\*\*

\*\*

```
For(int i=1; i<=n ; i++)  
{  
    cout << "**" << endl;  
}
```

$n = 6$

\*\*

\*\*

\*\*

\*\*

\*\*

\*\*

$n = 7$

\*\*

\*\*

\*\*

\*\*

\*\*

\*\*

\*\*

# 4 STARS

Given  $n$  (number of rows), print the following pattern:

$n = 5$

\*\*\*\*

\*\*\*\*

\*\*\*\*

\*\*\*\*

\*\*\*\*

$n = 6$

\*\*\*\*

\*\*\*\*

\*\*\*\*

\*\*\*\*

\*\*\*\*

\*\*\*\*

$n = 7$

\*\*\*\*

\*\*\*\*

\*\*\*\*

\*\*\*\*

\*\*\*\*

\*\*\*\*

\*\*\*\*

# M STARS (RECTANGLE)

Given  $n$  (no. of rows) and  $m$  (no. of cols) , print the following pattern:

$n = 5, m = 7$

\*\*\*\*\*

\*\*\*\*\*

\*\*\*\*\*

\*\*\*\*\*

\*\*\*\*\*

$n = 6, m = 3$

\*\*\*

\*\*\*

\*\*\*

\*\*\*

\*\*\*

\*\*\*

$n = 7, m = 4$

\*\*\*\*

\*\*\*\*

\*\*\*\*

\*\*\*\*

\*\*\*\*

\*\*\*\*

\*\*\*\*

# SQUARE

Given  $n$  (no. of rows and cols), print the following pattern:

$n = 5$

```
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *
```

$n = 6$

```
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *
```

$n = 7$

```
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *
```

# PYRAMID

Given  $n$  (no. of rows), print the following pattern:

$n = 5$

```
*  
**  
***  
****  
*****
```

$n = 6$

```
*  
**  
***  
****  
*****  
*****
```

$n = 7$

```
*  
**  
***  
****  
*****  
*****  
*****
```

# INVERTED PYRAMID

Given  $n$  (no. of rows), print the following pattern:

$n = 5$

\*\*\*\*\*

\*\*\*\*

\*\*\*

\*\*

\*

$n = 6$

\*\*\*\*\*

\*\*\*\*\*

\*\*\*\*

\*\*\*

\*\*

\*

$n = 7$

\*\*\*\*\*

\*\*\*\*\*

\*\*\*\*\*

\*\*\*\*

\*\*\*

\*\*

\*

# HOLLOW SQUARE

Boundary will exist

Given  $n$  (no. of rows and cols), print the following pattern:

$n = 5$

```
i=1 j=1
* * * * *
*
*
*
*
i=4 * * * * *
```

$n = 6$

```
* * * * *
*
*
*
*
*
* * * * *
```

# HOLLOW RECTANGLE

Given  $n$  (no. of rows) and  $m$  (no. of cols) , print the following pattern:

$n = 5, m = 7$

```
*****  
*      *  
*      *  
*      *  
*****
```

$n = 6, m = 8$

```
*****  
*      *  
*      *  
*      *  
*      *  
*****
```

$n = 7, m = 4$

```
****  
*  *  
*  *  
*  *  
*  *  
*  *  
****
```



# NUMBERED RECTANGLE

Given  $n$  (no. of rows) and  $m$  (no. of cols) , print the following pattern:

$n = 5, m = 7$

|         |               |
|---------|---------------|
| 1111111 | x x x x x x x |
| 2222222 | x x x x x x x |
| 3333333 | x x x x x x x |
| 4444444 | x x x x x x x |
| 5555555 | x x x x x x x |

$n = 6, m = 3$

|     |
|-----|
| 111 |
| 222 |
| 333 |
| 444 |
| 555 |
| 666 |

$n = 7, m = 4$

|         |
|---------|
| 1 1 1 1 |
| 2 2 2 2 |
| 3 3 3 3 |
| 4 4 4 4 |
| 5 5 5 5 |
| 6 6 6 6 |
| 7 7 7 7 |

- 1) Draw the star pattern
- 2) Replace  $x \rightarrow$  given elem

# NUMBERED RECTANGLE - II

Given  $n$  (no. of rows) and  $m$  (no. of cols) , print the following pattern:

$n = 5, m = 7$

1 2 3 4 5 6 7

1 2 3 4 5 6 7

1 2 3 4 5 6 7

1 2 3 4 5 6 7

1 2 3 4 5 6 7

$n = 6, m = 3$

1 2 3

1 2 3

1 2 3

1 2 3

1 2 3

1 2 3

$n = 7, m = 4$

1 2 3 4

1 2 3 4

1 2 3 4

1 2 3 4

1 2 3 4

1 2 3 4

1 2 3 4

# NUMBERED RECTANGLE - III

Given  $n$  (no. of rows) and  $m$  (no. of cols) , print the following pattern:

$n = 5, m = 7$

A B C D E F G

A B C D E F G

A B C D E F G

A B C D E F G

A B C D E F G

$n = 6, m = 3$

A B C

A B C

A B C

A B C

A B C

A B C

$n = 7, m = 4$

A B C D

A B C D

A B C D

A B C D

A B C D

A B C D

A B C D

# NUMBERED RECTANGLE – IV

Given  $n$  (no. of rows) and  $m$  (no. of cols) , print the following pattern:

$n = 5, m = 7$

```
AAAAAAA  
BBBBBBB  
CCCCCCC  
DDDDDDD  
EEEEEEE
```

$n = 6, m = 3$

```
AAA  
BBB  
CCC  
DDD  
EEE  
FFF
```

$n = 7, m = 4$

```
AAAA  
BBBB  
CCCC  
DDDD  
EEEE  
FFFF  
GGGG
```

# NUMBERED PYRAMID

Given  $n$  (no. of rows), print the following pattern:

$n = 5$

1

22

333

4444

55555

$n = 6$

1

22

333

4444

55555

666666

$n = 7$

1

22

333

4444

55555

666666

7777777

# BINARY PYRAMID

Given  $n$  (no. of rows), print the following pattern:

$n = 5$

```
0
10
010
1010
01010
```

$n = 6$

```
0
10
010
1010
01010
101010
```

$n = 7$

```
0
10
010
1010
01010
101010
0101010
```